

BSDS V: Regression Techniques

Problems Set 2: Least Squares Regression

Throughout this problem set, we will consider the multiple linear regression model with response variable y and a list of regressors/covariates $\{x_1, \dots, x_p\}$. Based on n observations the model is

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}, \quad \text{where} \quad \underline{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} 1 & x_{(1)} & \dots & x_{(p)} \end{bmatrix}, \quad \underline{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \text{and} \quad \underline{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}. \quad (1)$$

We assume that $E(\underline{\varepsilon}) = \underline{0}$ and $\text{var}(\underline{\varepsilon}) = \sigma^2 I_n$.

A. Problems without any distributional assumption on $\underline{\varepsilon}$.

1. Consider the least square solution of $\underline{\beta}$, say $\hat{\underline{\beta}}_{\text{OLS}}$, obtained from the linear regression model in (1), where the covariates are centered, i.e., $\underline{1}^\top \underline{x}_{(j)} = 0$ for all $j = 1, \dots, p$. Further, consider the j -th simple linear regression model

$$Y_i = \beta_0 + \beta_j x_{i,j} + \varepsilon_i, \quad i = 1, \dots, n,$$

and let $\hat{\beta}_{j,S}$ be the solution to the simple linear regression model, and $\hat{\beta}_S = \begin{bmatrix} \bar{Y} & \hat{\beta}_{1,S} & \dots & \hat{\beta}_{p,S} \end{bmatrix}^\top$.

- (a) Compare $\hat{\underline{\beta}}_{\text{OLS}}$ and $\hat{\beta}_S$.
 - (b) When will these two solutions be equal?
2. Consider the regression model in (1), and let the design matrix satisfy the following orthogonality property:

$$\underline{1}^\top \underline{x}_{(j)} = 0, \quad \text{and} \quad \underline{x}_{(k)}^\top \underline{x}_{(j)} = 0, \quad \text{for all } k \neq j, \quad j, k = 1, \dots, p.$$

- (a) Find the expression of $\hat{\underline{\beta}}_{\text{OLS}}$ under this situation.
 - (b) Find the expression of the residual sum of squares (RSS) under this situation.
3. Consider the regression model in (1), and let the design matrix satisfy the following orthogonality property:

$$\underline{1}^\top \underline{x}_{(j)} = 0, \quad \text{and} \quad \|\underline{x}_{(j)}\| = c, \quad \text{for all } j = 1, \dots, p.$$

Show that $\sum_{j=1}^p \text{var}(\hat{\beta}_{j,\text{OLS}})$ is minimized when the covariates are orthogonal (as in Q.2).

4. Let the design matrix X has rank $p + 1$. Then show that, for any parameter of the form $\underline{c}^\top \underline{\beta}$, the estimator $\underline{c}^\top \hat{\underline{\beta}}_{\text{OLS}}$ is the best linear unbiased estimator.
5. Let

$$Y_i = \beta_0 + \beta_1(x_{i1} - \bar{x}_1) + \beta_2(x_{i2} - \bar{x}_2) + \varepsilon_i, \quad (i = 1, 2, \dots, n),$$

where

$$\bar{x}_j = \sum_{i=1}^n \frac{x_{ij}}{n}, \quad E[\varepsilon] = 0, \quad \text{Var}[\varepsilon] = \sigma^2 I_n.$$

If $\hat{\beta}_1$ is the least squares estimate of β_1 , show that

$$\text{Var}[\hat{\beta}_1] = \frac{\sigma^2}{\sum_i (x_{i1} - \bar{x}_1)^2 (1 - r^2)},$$

where r is the correlation coefficient of the n pairs (x_{i1}, x_{i2}) .

6. Let $Y_1, Y_2, \dots, Y_n$ be n independent random variables with common variance σ^2 and common third and fourth moments, μ_3 and μ_4 , respectively, about their means. If $E[\mathbf{Y}] = X\beta$, where X is an $n \times p$ matrix of rank p , then $(n - p)S^2$ is the unique nonnegative quadratic unbiased estimate of $(n - p)\sigma^2$ with minimum variance when $\mu_4 = 3\sigma^4$ or when the diagonal elements of P are all equal.

[Hint: See Seber and Lee, Chapter 3, Section 3.3]

B. Problems with normality assumption on ε_i . Suppose in addition to the regression model in (1) we assume that $\varepsilon_i \stackrel{i.i.d.}{\sim} N(0, 1)$.

1. Show that $\hat{\beta}_{\text{OLS}}$ and RSS are independent.
2. Show that both $\hat{\beta}_{\text{ML}}$ and $\hat{\sigma}_{\text{ML}}^2$ are consistent estimators.
3. Under the normal linear regression model described above, consider the following problems of testing:
 - (a) Suppose there is a hierarchy among the regressors, so that we want to test $H_0 : \beta_1 - \beta_2 = \dots = \beta_{p-1} - \beta_p = 1$. Find the LRT test statistic for this testing explicitly and its exact distribution.
 - (b) Suppose we want to test $H_0 : \beta_1 + \beta_2 + \beta_p = k$. Find the LRT test statistic for this testing explicitly and its exact distribution.
4. Consider the simple linear regression model $Y_i = \alpha + \beta x + \varepsilon_i$, for $i = 1, \dots, n$, and the problem of testing $H_0 : \beta = 0$. Let $\Lambda(\underline{Y})$ is the LRT test statistic for testing H_0 against H_1 : no restriction. Show that $r^2 = 1 - \{\Lambda(\underline{Y})\}^{2/n}$.

[This, along with Result 6 & 7 in the classnotes, shows that the simple correlation coefficient (SCC), multiple correlation coefficient (MCC) and partial correlation coefficient (PCC) share same relation with the LRT test statistics corresponding to the tests $H_{0S} : \beta = 0$ (in simple regression setup), $H_{0M} : \beta_1 = \dots = \beta_p = 0$ and $H_{0P} : \beta_p = 0$ (last two in multiple regression setup).]

5. In the normal linear regression model as described above, consider the problem of finding the simultaneous confidence intervals (SCIs) for $\beta_1, \beta_2, \beta_1 + \beta_2$ and $\beta_1 - \beta_2$. Find Scheffe's SCI for the confidence coefficient $1 - \alpha$.
6. Consider a linear combination of $\underline{\beta}$, say $\underline{a}^\top \underline{\beta}$. Find the Scheffe's confidence interval of the form $[\underline{a}^\top \hat{\beta}_{\text{OLS}} \pm G_\alpha(\underline{a})]$ which is valid for all $\underline{a} \neq 0$. In particular, consider a given value of $\underline{a} = \underline{a}_0$. Find the individual confidence interval for $\underline{a}_0^\top \underline{\beta}$ of the form $[\underline{a}_0^\top \hat{\beta}_{\text{OLS}} \pm H_\alpha(\underline{a}_0)]$. Compare $G_\alpha(\underline{a}_0)$ and $H_\alpha(\underline{a}_0)$. Which confidence interval is wider and why?
7. Consider the problem of finding confidence intervals for two new observations $\underline{x}_0$ (known) and $\underline{z}_0$ (known), where $\underline{x}_0$ and $\underline{z}_0$ are unit eigenvectors corresponding to the highest and smallest eigenvalues of $X^\top X$, say $\ell_{\max}$ and $\ell_{\min}$ respectively, and $\ell_{\max}/\ell_{\min} = 10^4$. Interpret the result. Compare the widths of confidence intervals of $\underline{x}_0^\top \underline{\beta}$ and $\underline{z}_0^\top \underline{\beta}$.

C. Practical problems. Consider the Boston Housing (BH) dataset and the Iris dataset.

For the **Boston Housing** dataset, consider *MEDV* as the response variable and *CRIM*, *ZN*, *INDUS*, *NOX*, *RM*, *AGE*, *DIS*, *RAD*, *TAX*, *PTRATIO*, *B*, and *LSTAT* as the explanatory (regressor) variables.

For the **Iris** dataset, consider *Sepal Length* as the response variable and *Sepal Width*, *Petal Length*, and *Petal Width* as the explanatory (regressor) variables.

For each dataset, carry out the following analyses:

1. Consider separate simple linear regression models, with the response variable regressed on each explanatory variable individually. For each model, test whether the corresponding regressor has a statistically significant effect on the response.
2. Consider a multiple linear regression model in which the response variable is regressed simultaneously on all the explanatory variables. Perform the following two types of hypothesis tests:
 - (a) For each regressor individually, test whether it has a statistically significant effect on the response after accounting for the effects of all the other regressors. In other words, for a fixed $j = 1, \dots, p$, test $H_0 : \beta_j = 0$ against the appropriate alternative, with no restrictions imposed on β_k for $k \neq j$.
 - (b) Test whether the regressors, taken together, have a statistically significant effect on the response. That is, test $H_0 : \underline{\beta} = \underline{0}$, where $\underline{\beta} = (\beta_1, \dots, \beta_p)^\top$. Find the coefficient of determination, and conclude on the strength of association of the response and the covariates.
 - (c) Consider the problem of testing that all regressors (except the intercept) have same effect on the response. What is the estimate of the common regression coefficient β and the associated p -value?
3. For each regressor, compare the regression coefficient and corresponding p -value obtained from:
 - (a) the simple linear regression model,
 - (b) the multiple linear regression model with the individual test of $H_0 : \beta_j = 0$ (i.e., the test in 2(a)).

Based on the testing results, answer the following questions:

- (a) Are the regression coefficients have same sign in both cases? If not, explain the possible reasons in view of the underlying dataset.
 - (b) Are the resulting p -values of the same order of magnitude? If they differ substantially, explain the possible reasons for the differences by examining the relationships among the variables in the dataset.
4. Standardize the data by centering and scaling each explanatory variable. Perform the testing in Q.2(c). Are the results you obtain now same as those prior to standardization? If there are any changes, explain why. In particular, discuss the role of standardization and the scale of measurement of the variables in the context of the two datasets.
5. Consider drawing an ellipse from the equation $(\underline{\beta} - \widehat{\underline{\beta}})^\top L (\underline{\beta} - \widehat{\underline{\beta}}) = c$, where $\underline{\beta} = \begin{bmatrix} \beta_1 \\ \beta_2 \end{bmatrix} \in \mathbb{R}^2$, $\widehat{\underline{\beta}} = (2, 3)^\top$ $L = ((1 - 0.5^{|i-j|}))$ is symmetric p.d., $i, j \in \{1, 2\}$ and $c = 5$.
 - (a) Find the transformation $\beta_1 \rightarrow X$ and $\beta_2 \rightarrow Y$, and constants (a, b) such that the above equation of the ellipse can be written as $X^2/a^2 + Y^2/b^2 = c$.
 - (b) Show that with respect to $(X/a, Y/b)$ -coordinate system, the above equation represents a circle with center $(0, 0)$ and radius $\sqrt{c}$.
 - (c) Hence, show that with respect to (X, Y) -coordinate system, the above equation represents a standard ellipse (where the two axes are the natural axes) with center $(0, 0)$ and (a, b) as the major and minor axes.

- (d) Hence, show that with respect to $(\beta_1 - \hat{\beta}_1, \beta_2 - \hat{\beta}_2)$ coordinate system the above equation represents a non-standard ellipse with center $(0, 0)$.
- (e) Hence, show that with respect to (β_1, β_2) coordinate system the above equation represents a non-standard ellipse with center $(\hat{\beta}_1, \hat{\beta}_1)$ with same angle and axes as in (d).

6. Consider a data set with $n = 20$ observations on (y, x_1, x_2) with the following specifications:

$$\begin{aligned} \sum_i x_{1,i} &= 1.8; & \sum_i x_{1,i}^2 &= 21.2; & \sum_i x_{2,i} &= 14.5; & \sum_i x_{2,i}^2 &= 83.7; \\ \sum_i y_i &= -66.6; & \sum_i x_{1,i}x_{2,i} &= -4; & \sum_i x_{1,i}y_i &= 34.6; & \sum_i x_{2,i}y_i &= -127.1. \end{aligned}$$

Further, if we fit a normal linear regression model $y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \varepsilon_i$, $i = 1, \dots, 20$, then the RSS would be 576.2. Suppose we are interested in finding simultaneous confidence interval for (β_1, β_2) . In the same plot, draw the following choices of simultaneous confidence intervals (SCI):

- (a) The SCI obtained by the smallest rectangle containing the confidence ellipsoid. (*Note that, in this case the confidence ellipsoid is approximately of the form $X^2/a^2 + Y^2/b^2 = c^2$ after centering. So smallest rectangle can be easily obtained.*)
- (b) The SCI obtained by Bonferroni's method.
- (c) The SCI obtained by Scheffe's method.

Which of these three methods provide the best result?